DBMS Project

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Project Title: Financial Market Anomaly & Crash Detection System

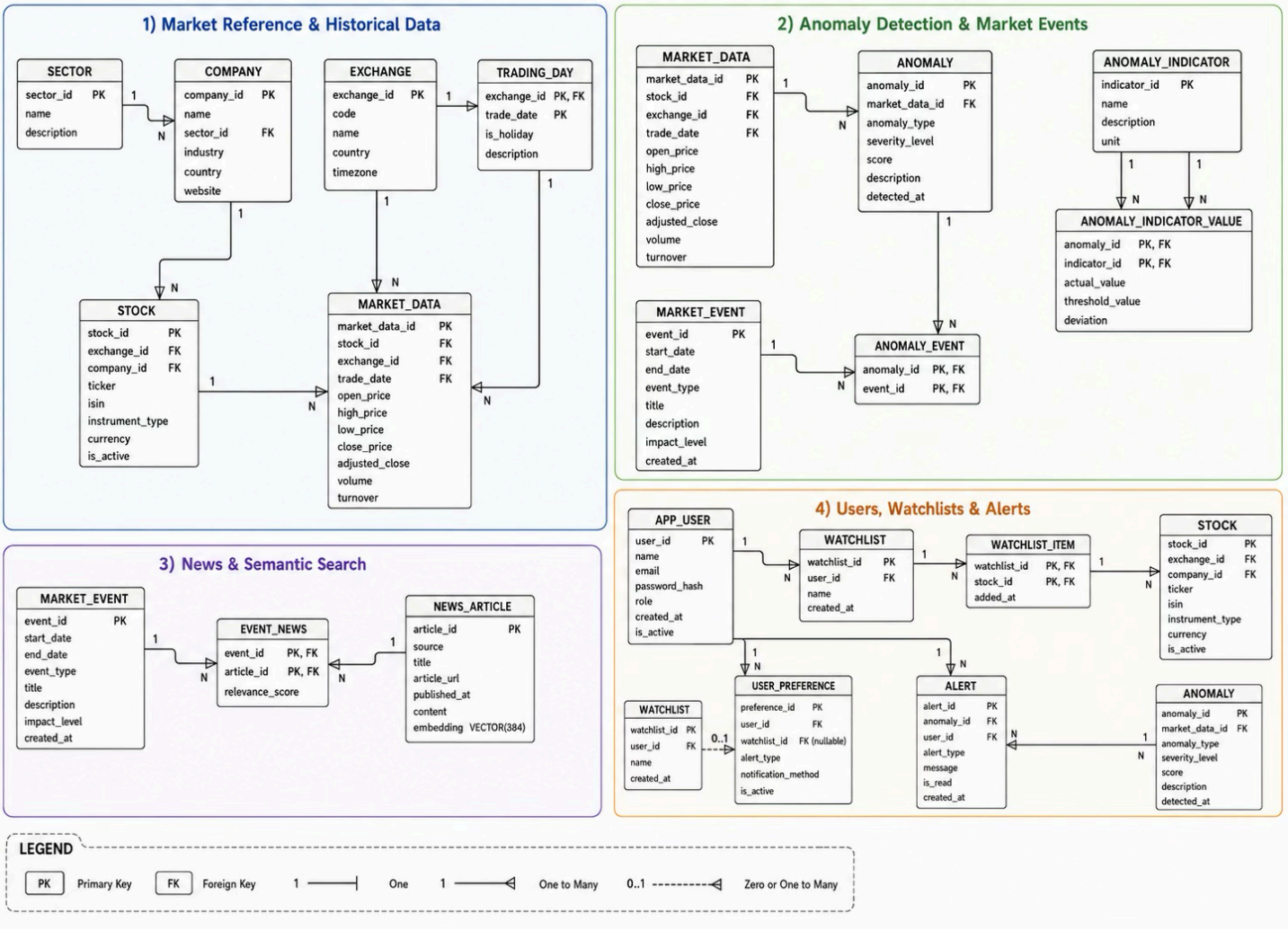
Problem Statement: Financial markets generate large volumes of structured and unstructured data, including prices, trading volumes, volatility measurements, and financial news. Identifying unusual market behaviour manually is difficult because significant events may involve simultaneous changes across multiple securities and sectors. Existing basic market dashboards primarily display historical/current prices but do not provide an integrated system for detecting anomalies and investigating similar historical events.
Our system will address this by combining relational financial data with anomaly detection and semantic analysis of market-related news/events.

Database Technology Used: PostgreSQL 17.11 is used as the primary relational database management system for the project. PostgreSQL was selected because it provides strong support for relational data modelling, primary and foreign key constraints, transactions, indexing, and complex queries required for managing financial market data.

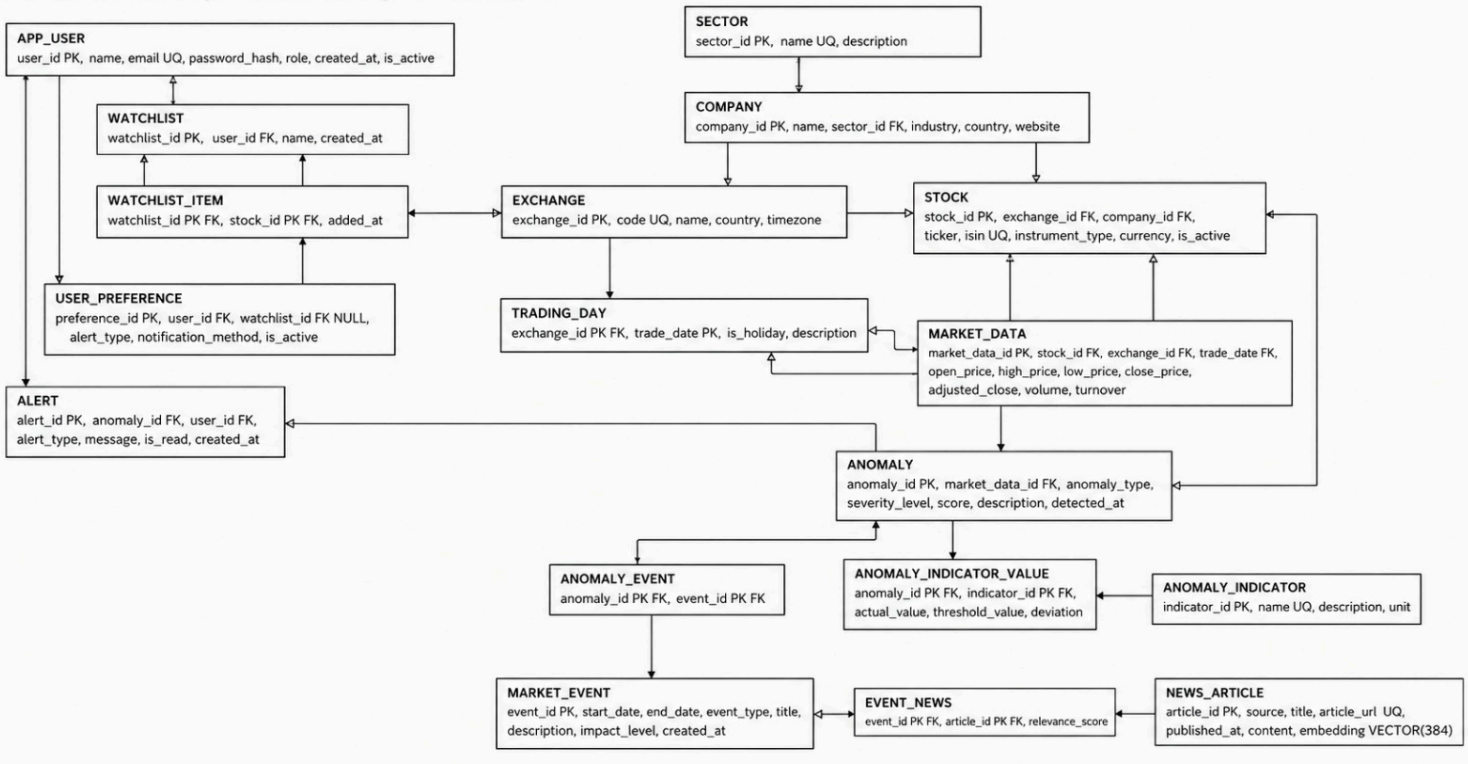
The database is designed using a normalized relational schema consisting of 18 tables covering market reference data, historical market data, anomalies, market events, news articles, users, watchlists, and alerts.

The project also uses the pgvector extension (v0.8.6) in PostgreSQL to store 384-dimensional vector embeddings of news articles. This enables semantic similarity search between financial news articles and detected market events/anomalies. An HNSW vector index with cosine distance is used to support efficient similarity search.

Financial Market Anomaly & Crash Detection System — ER Diagram



Financial Market Anomaly & Crash Detection System – Relational Schema



DDL Implementation

Team Project — -zsh — 100×62

```
Referenced by:
    TABLE "market_data" CONSTRAINT "fk_market_data_stock" FOREIGN KEY (stock_id) REFERENCES stock(st
ock_id)
    TABLE "watchlist_item" CONSTRAINT "fk_watchlist_item_stock" FOREIGN KEY (stock_id) REFERENCES st
ock(stock_id) ON DELETE CASCADE

[amritraj@Amrits-Laptop Team Project % psql -d financial_market_db -c "\di"                         ]
                            List of relations
 Schema |             Name              | Type  |  Owner   |          Table
--------+-------------------------------+-------+----------+-------------------------
 public | alert_pkey                    | index | amritraj | alert
 public | anomaly_indicator_name_key    | index | amritraj | anomaly_indicator
 public | anomaly_indicator_pkey        | index | amritraj | anomaly_indicator
 public | anomaly_pkey                  | index | amritraj | anomaly
 public | app_user_email_key            | index | amritraj | app_user
 public | app_user_pkey                 | index | amritraj | app_user
 public | company_pkey                  | index | amritraj | company
 public | exchange_code_key             | index | amritraj | exchange
 public | exchange_pkey                 | index | amritraj | exchange
 public | idx_alert_anomaly             | index | amritraj | alert
 public | idx_alert_user                | index | amritraj | alert
 public | idx_anomaly_detected_at       | index | amritraj | anomaly
 public | idx_anomaly_market_data       | index | amritraj | anomaly
 public | idx_anomaly_severity          | index | amritraj | anomaly
 public | idx_anomaly_type              | index | amritraj | anomaly
 public | idx_company_sector            | index | amritraj | company
 public | idx_market_data_stock_date    | index | amritraj | market_data
 public | idx_market_data_trade_date    | index | amritraj | market_data
 public | idx_market_event_dates        | index | amritraj | market_event
 public | idx_market_event_type         | index | amritraj | market_event
 public | idx_news_embedding            | index | amritraj | news_article
 public | idx_news_published_at         | index | amritraj | news_article
 public | idx_news_source               | index | amritraj | news_article
 public | idx_stock_company             | index | amritraj | stock
 public | idx_stock_exchange            | index | amritraj | stock
 public | idx_trading_day_date          | index | amritraj | trading_day
 public | idx_watchlist_item_stock      | index | amritraj | watchlist_item
 public | idx_watchlist_user            | index | amritraj | watchlist
 public | market_data_pkey              | index | amritraj | market_data
 public | market_event_pkey             | index | amritraj | market_event
 public | news_article_article_url_key  | index | amritraj | news_article
 public | news_article_pkey             | index | amritraj | news_article
 public | pk_anomaly_event              | index | amritraj | anomaly_event
 public | pk_anomaly_indicator_value    | index | amritraj | anomaly_indicator_value
 public | pk_event_news                 | index | amritraj | event_news
 public | pk_trading_day                | index | amritraj | trading_day
 public | pk_watchlist_item             | index | amritraj | watchlist_item
 public | sector_name_key               | index | amritraj | sector
 public | sector_pkey                   | index | amritraj | sector
 public | stock_isin_key                | index | amritraj | stock
 public | stock_pkey                    | index | amritraj | stock
 public | uq_market_data_stock_date     | index | amritraj | market_data
 public | uq_stock_exchange_ticker      | index | amritraj | stock
 public | uq_watchlist_user_name        | index | amritraj | watchlist
 public | user_preference_pkey          | index | amritraj | user_preference
 public | watchlist_pkey                | index | amritraj | watchlist
(46 rows)
```

```
[amritraj@Amrits-Laptop Team Project % psql -d financial_market_db -c "\dv"                    ]
 Did not find any relations.
[amritraj@Amrits-Laptop Team Project % clear                                                    ]

[amritraj@Amrits-Laptop Team Project % createdb financial_market_ddl_demo                      ]
[amritraj@Amrits-Laptop Team Project % psql -d financial_market_ddl_demo -c "CREATE EXTENSION IF NOT]
  EXISTS vector;"
 CREATE EXTENSION
[amritraj@Amrits-Laptop Team Project % psql -d financial_market_ddl_demo -v ON_ERROR_STOP=1 -f datab]
 ase/ddl/financial_market_schema.sql
 CREATE TABLE
 CREATE TABLE
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 CREATE INDEX
```

```
[amritraj@Amrits-Laptop Team Project % createdb financial_market_ddl_commands                    ]
[amritraj@Amrits-Laptop Team Project % psql -d financial_market_ddl_commands -c "CREATE EXTENSION IF]
 NOT EXISTS vector;"
CREATE EXTENSION
[amritraj@Amrits-Laptop Team Project % psql -d financial_market_ddl_commands -a -v ON_ERROR_STOP=1 -]
f database/ddl/financial_market_schema.sql
-- Financial Market Anomaly & Crash Detection System
-- Database DDL
-- PostgreSQL 17 + pgvector
-- ============================================================
-- 1. MARKET & REFERENCE DATA
-- ============================================================
-- ------------------------------------------------------------
-- SECTOR
-- ------------------------------------------------------------
CREATE TABLE sector (
    sector_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL UNIQUE,
    description TEXT
);
CREATE TABLE
-- ------------------------------------------------------------
-- COMPANY
-- ------------------------------------------------------------
CREATE TABLE company (
    company_id SERIAL PRIMARY KEY,
    name VARCHAR(150) NOT NULL,
    sector_id INT NOT NULL,
    industry VARCHAR(100),
    country VARCHAR(100),
    website VARCHAR(255),
    CONSTRAINT fk_company_sector
        FOREIGN KEY (sector_id)
        REFERENCES sector(sector_id)
);
CREATE TABLE
-- ------------------------------------------------------------
-- EXCHANGE
-- ------------------------------------------------------------
CREATE TABLE exchange (
    exchange_id SERIAL PRIMARY KEY,
    code VARCHAR(20) NOT NULL UNIQUE,
    name VARCHAR(100) NOT NULL,
    country VARCHAR(100),
    timezone VARCHAR(50)
);
CREATE TABLE
-- ------------------------------------------------------------
-- STOCK
-- ------------------------------------------------------------
CREATE TABLE stock (
    stock_id SERIAL PRIMARY KEY,
    exchange_id INT NOT NULL,
    company_id INT NOT NULL,
    ticker VARCHAR(20) NOT NULL,
    isin VARCHAR(20) UNIQUE,
    instrument_type VARCHAR(50),
    currency VARCHAR(10),
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    CONSTRAINT fk_stock_exchange
        FOREIGN KEY (exchange_id)
```

```
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    CONSTRAINT fk_stock_exchange
        FOREIGN KEY (exchange_id)
        REFERENCES exchange(exchange_id),
    CONSTRAINT fk_stock_company
        FOREIGN KEY (company_id)
        REFERENCES company(company_id),
    CONSTRAINT uq_stock_exchange_ticker
        UNIQUE (exchange_id, ticker)
);
CREATE TABLE
-- ------------------------------------------------------------
-- TRADING_DAY
-- ------------------------------------------------------------
CREATE TABLE trading_day (
    exchange_id INT NOT NULL,
    trade_date DATE NOT NULL,
    is_holiday BOOLEAN NOT NULL DEFAULT FALSE,
    description TEXT,
    CONSTRAINT pk_trading_day
        PRIMARY KEY (exchange_id, trade_date),
    CONSTRAINT fk_trading_day_exchange
        FOREIGN KEY (exchange_id)
        REFERENCES exchange(exchange_id)
);
CREATE TABLE
-- ------------------------------------------------------------
-- MARKET_DATA
-- ------------------------------------------------------------
CREATE TABLE market_data (
    market_data_id BIGSERIAL PRIMARY KEY,
    stock_id INT NOT NULL,
    exchange_id INT NOT NULL,
    trade_date DATE NOT NULL,
    open_price NUMERIC(18,6),
    high_price NUMERIC(18,6),
    low_price NUMERIC(18,6),
    close_price NUMERIC(18,6),
    adjusted_close NUMERIC(18,6),
    volume BIGINT,
    turnover NUMERIC(20,4),
    CONSTRAINT fk_market_data_stock
        FOREIGN KEY (stock_id)
        REFERENCES stock(stock_id),
    CONSTRAINT fk_market_data_trading_day
        FOREIGN KEY (exchange_id, trade_date)
        REFERENCES trading_day(exchange_id, trade_date),
    CONSTRAINT uq_market_data_stock_date
        UNIQUE (stock_id, trade_date),
    CONSTRAINT chk_market_data_prices
        CHECK (
            open_price IS NULL OR open_price >= 0
        ),
    CONSTRAINT chk_market_data_high
        CHECK (
            high_price IS NULL OR high_price >= 0
        ),
    CONSTRAINT chk_market_data_low
        CHECK (
            low_price IS NULL OR low_price >= 0
        ),
    CONSTRAINT chk_market_data_close
```

```
        REFERENCES trading_day(exchange_id, trade_date),
    CONSTRAINT uq_market_data_stock_date
        UNIQUE (stock_id, trade_date),
    CONSTRAINT chk_market_data_prices
        CHECK (
            open_price IS NULL OR open_price >= 0
        ),
    CONSTRAINT chk_market_data_high
        CHECK (
            high_price IS NULL OR high_price >= 0
        ),
    CONSTRAINT chk_market_data_low
        CHECK (
            low_price IS NULL OR low_price >= 0
        ),
    CONSTRAINT chk_market_data_close
        CHECK (
            close_price IS NULL OR close_price >= 0
        ),
    CONSTRAINT chk_market_data_adjusted_close
        CHECK (
            adjusted_close IS NULL OR adjusted_close >= 0
        ),
    CONSTRAINT chk_market_data_volume
        CHECK (
            volume IS NULL OR volume >= 0
        ),
    CONSTRAINT chk_market_data_turnover
        CHECK (
            turnover IS NULL OR turnover >= 0
        ),
    CONSTRAINT chk_market_data_high_low
        CHECK (
            high_price IS NULL
            OR low_price IS NULL
            OR high_price >= low_price
        )
);
CREATE TABLE
-- ============================================================
-- 2. ANOMALY, EVENT & NEWS DATA
-- ============================================================
-- ------------------------------------------------------------
-- ANOMALY_INDICATOR
-- ------------------------------------------------------------
CREATE TABLE anomaly_indicator (
    indicator_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL UNIQUE,
    description TEXT,
    unit VARCHAR(50)
);
CREATE TABLE
-- ------------------------------------------------------------
-- ANOMALY
-- ------------------------------------------------------------
CREATE TABLE anomaly (
    anomaly_id BIGSERIAL PRIMARY KEY,
    market_data_id BIGINT NOT NULL,
    anomaly_type VARCHAR(100) NOT NULL,
    severity_level VARCHAR(20) NOT NULL,
    score NUMERIC(10,4),
    description TEXT,
```

```
    score NUMERIC(10,4),
    description TEXT,
    detected_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_anomaly_market_data
        FOREIGN KEY (market_data_id)
        REFERENCES market_data(market_data_id),
    CONSTRAINT chk_anomaly_score
        CHECK (
            score IS NULL OR score >= 0
        ),
    CONSTRAINT chk_anomaly_severity
        CHECK (
            severity_level IN ('LOW', 'MEDIUM', 'HIGH', 'CRITICAL')
        )
);
CREATE TABLE
-- ------------------------------------------------------------
-- ANOMALY_INDICATOR_VALUE
-- ------------------------------------------------------------
CREATE TABLE anomaly_indicator_value (
    anomaly_id BIGINT NOT NULL,
    indicator_id INT NOT NULL,
    actual_value NUMERIC(20,8),
    threshold_value NUMERIC(20,8),
    deviation NUMERIC(20,8),
    CONSTRAINT pk_anomaly_indicator_value
        PRIMARY KEY (anomaly_id, indicator_id),
    CONSTRAINT fk_aiv_anomaly
        FOREIGN KEY (anomaly_id)
        REFERENCES anomaly(anomaly_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_aiv_indicator
        FOREIGN KEY (indicator_id)
        REFERENCES anomaly_indicator(indicator_id)
);
CREATE TABLE
-- ------------------------------------------------------------
-- MARKET_EVENT
-- ------------------------------------------------------------
CREATE TABLE market_event (
    event_id BIGSERIAL PRIMARY KEY,
    start_date DATE NOT NULL,
    end_date DATE,
    event_type VARCHAR(100) NOT NULL,
    title VARCHAR(255) NOT NULL,
    description TEXT,
    impact_level VARCHAR(20),
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_market_event_dates
        CHECK (
            end_date IS NULL OR end_date >= start_date
        ),
    CONSTRAINT chk_market_event_impact
        CHECK (
            impact_level IS NULL
            OR impact_level IN ('LOW', 'MEDIUM', 'HIGH', 'CRITICAL')
        )
);
CREATE TABLE
-- ------------------------------------------------------------
-- ANOMALY_EVENT
-- Many-to-many relationship:
```

```
);
CREATE TABLE
-- ------------------------------------------------------------
-- ANOMALY_EVENT
-- Many-to-many relationship:
-- ANOMALY <-> MARKET_EVENT
-- ------------------------------------------------------------
CREATE TABLE anomaly_event (
    anomaly_id BIGINT NOT NULL,
    event_id BIGINT NOT NULL,
    CONSTRAINT pk_anomaly_event
        PRIMARY KEY (anomaly_id, event_id),
    CONSTRAINT fk_anomaly_event_anomaly
        FOREIGN KEY (anomaly_id)
        REFERENCES anomaly(anomaly_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_anomaly_event_event
        FOREIGN KEY (event_id)
        REFERENCES market_event(event_id)
        ON DELETE CASCADE
);
CREATE TABLE
-- ------------------------------------------------------------
-- NEWS_ARTICLE
-- ------------------------------------------------------------
CREATE TABLE news_article (
    article_id BIGSERIAL PRIMARY KEY,
    source VARCHAR(150) NOT NULL,
    title VARCHAR(500) NOT NULL,
    article_url VARCHAR(1000) UNIQUE,
    published_at TIMESTAMP,
    content TEXT,
    -- 384-dimensional embedding for semantic search
    embedding VECTOR(384)
);
CREATE TABLE
-- ------------------------------------------------------------
-- EVENT_NEWS
-- Many-to-many relationship:
-- MARKET_EVENT <-> NEWS_ARTICLE
-- ------------------------------------------------------------
CREATE TABLE event_news (
    event_id BIGINT NOT NULL,
    article_id BIGINT NOT NULL,
    relevance_score NUMERIC(6,4),
    CONSTRAINT pk_event_news
        PRIMARY KEY (event_id, article_id),
    CONSTRAINT fk_event_news_event
        FOREIGN KEY (event_id)
        REFERENCES market_event(event_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_event_news_article
        FOREIGN KEY (article_id)
        REFERENCES news_article(article_id)
        ON DELETE CASCADE,
    CONSTRAINT chk_event_news_relevance
        CHECK (
            relevance_score IS NULL
            OR (relevance_score >= 0 AND relevance_score <= 1)
        )
);
CREATE TABLE
```

```
CREATE TABLE
-- ============================================================
-- 3. USERS, WATCHLISTS & ALERTS
-- ============================================================
-- ------------------------------------------------------------
-- APP_USER
-- ------------------------------------------------------------
CREATE TABLE app_user (
    user_id BIGSERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(255) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    role VARCHAR(30) NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    is_active BOOLEAN NOT NULL DEFAULT TRUE
);
CREATE TABLE
-- ------------------------------------------------------------
-- WATCHLIST
-- ------------------------------------------------------------
CREATE TABLE watchlist (
    watchlist_id BIGSERIAL PRIMARY KEY,
    user_id BIGINT NOT NULL,
    name VARCHAR(100) NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_watchlist_user
        FOREIGN KEY (user_id)
        REFERENCES app_user(user_id)
        ON DELETE CASCADE,
    CONSTRAINT uq_watchlist_user_name
        UNIQUE (user_id, name)
);
CREATE TABLE
-- ------------------------------------------------------------
-- WATCHLIST_ITEM
-- Many-to-many relationship:
-- WATCHLIST <-> STOCK
-- ------------------------------------------------------------
CREATE TABLE watchlist_item (
    watchlist_id BIGINT NOT NULL,
    stock_id INT NOT NULL,
    added_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_watchlist_item
        PRIMARY KEY (watchlist_id, stock_id),
    CONSTRAINT fk_watchlist_item_watchlist
        FOREIGN KEY (watchlist_id)
        REFERENCES watchlist(watchlist_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_watchlist_item_stock
        FOREIGN KEY (stock_id)
        REFERENCES stock(stock_id)
        ON DELETE CASCADE
);
CREATE TABLE
-- ------------------------------------------------------------
-- USER_PREFERENCE
-- ------------------------------------------------------------
CREATE TABLE user_preference (
    preference_id BIGSERIAL PRIMARY KEY,
    user_id BIGINT NOT NULL,
    watchlist_id BIGINT,
    alert_type VARCHAR(50),
```

```
CREATE TABLE user_preference (
    preference_id BIGSERIAL PRIMARY KEY,
    user_id BIGINT NOT NULL,
    watchlist_id BIGINT,
    alert_type VARCHAR(50),
    notification_method VARCHAR(50),
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    CONSTRAINT fk_user_preference_user
        FOREIGN KEY (user_id)
        REFERENCES app_user(user_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_user_preference_watchlist
        FOREIGN KEY (watchlist_id)
        REFERENCES watchlist(watchlist_id)
        ON DELETE SET NULL
);
CREATE TABLE
-- ------------------------------------------------------------
-- ALERT
-- ------------------------------------------------------------
CREATE TABLE alert (
    alert_id BIGSERIAL PRIMARY KEY,
    anomaly_id BIGINT NOT NULL,
    user_id BIGINT NOT NULL,
    alert_type VARCHAR(50),
    message TEXT,
    is_read BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_alert_anomaly
        FOREIGN KEY (anomaly_id)
        REFERENCES anomaly(anomaly_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_alert_user
        FOREIGN KEY (user_id)
        REFERENCES app_user(user_id)
        ON DELETE CASCADE
);
CREATE TABLE
-- ============================================================
-- 4. INDEXES
-- ============================================================
-- Company / stock lookups
CREATE INDEX idx_company_sector
    ON company(sector_id);
CREATE INDEX
CREATE INDEX idx_stock_company
    ON stock(company_id);
CREATE INDEX
CREATE INDEX idx_stock_exchange
    ON stock(exchange_id);
CREATE INDEX
-- Trading / market data
CREATE INDEX idx_trading_day_date
    ON trading_day(trade_date);
CREATE INDEX
CREATE INDEX idx_market_data_stock_date
    ON market_data(stock_id, trade_date);
CREATE INDEX
CREATE INDEX idx_market_data_trade_date
    ON market_data(trade_date);
CREATE INDEX
-- Anomaly queries
```

```
-- Company / stock lookups
CREATE INDEX idx_company_sector
    ON company(sector_id);
CREATE INDEX
CREATE INDEX idx_stock_company
    ON stock(company_id);
CREATE INDEX
CREATE INDEX idx_stock_exchange
    ON stock(exchange_id);
CREATE INDEX
-- Trading / market data
CREATE INDEX idx_trading_day_date
    ON trading_day(trade_date);
CREATE INDEX
CREATE INDEX idx_market_data_stock_date
    ON market_data(stock_id, trade_date);
CREATE INDEX
CREATE INDEX idx_market_data_trade_date
    ON market_data(trade_date);
CREATE INDEX
-- Anomaly queries
CREATE INDEX idx_anomaly_market_data
    ON anomaly(market_data_id);
CREATE INDEX
CREATE INDEX idx_anomaly_type
    ON anomaly(anomaly_type);
CREATE INDEX
CREATE INDEX idx_anomaly_severity
    ON anomaly(severity_level);
CREATE INDEX
CREATE INDEX idx_anomaly_detected_at
    ON anomaly(detected_at);
CREATE INDEX
-- Events
CREATE INDEX idx_market_event_dates
    ON market_event(start_date, end_date);
CREATE INDEX
CREATE INDEX idx_market_event_type
    ON market_event(event_type);
CREATE INDEX
-- News
CREATE INDEX idx_news_published_at
    ON news_article(published_at);
CREATE INDEX
CREATE INDEX idx_news_source
    ON news_article(source);
CREATE INDEX
-- Users / alerts
CREATE INDEX idx_watchlist_user
    ON watchlist(user_id);
CREATE INDEX
CREATE INDEX idx_watchlist_item_stock
    ON watchlist_item(stock_id);
CREATE INDEX
CREATE INDEX idx_alert_user
    ON alert(user_id);
CREATE INDEX
CREATE INDEX idx_alert_anomaly
    ON alert(anomaly_id);
CREATE INDEX
-- ============================================================
-- 5. VECTOR SIMILARITY INDEX
```



```
CREATE INDEX idx_market_event_type
    ON market_event(event_type);
CREATE INDEX
-- News
CREATE INDEX idx_news_published_at
    ON news_article(published_at);
CREATE INDEX
CREATE INDEX idx_news_source
    ON news_article(source);
CREATE INDEX
-- Users / alerts
CREATE INDEX idx_watchlist_user
    ON watchlist(user_id);
CREATE INDEX
CREATE INDEX idx_watchlist_item_stock
    ON watchlist_item(stock_id);
CREATE INDEX
CREATE INDEX idx_alert_user
    ON alert(user_id);
CREATE INDEX
CREATE INDEX idx_alert_anomaly
    ON alert(anomaly_id);
CREATE INDEX
-- ============================================================
-- 5. VECTOR SIMILARITY INDEX
-- ============================================================
CREATE INDEX idx_news_embedding
    ON news_article
    USING hnsw (embedding vector_cosine_ops);
CREATE INDEX
-- ============================================================
-- END OF DDL
-- ============================================================
amritraj@Amrits-Laptop Team Project % psql financial_market_ddl_commands
psql (17.11 (Homebrew))
Type "help" for help.

financial_market_ddl_commands=# \pset pager off
Pager usage is off.
financial_market_ddl_commands=# \dt
                List of relations
 Schema |          Name           | Type  |  Owner
--------+-------------------------+-------+----------
 public | alert                   | table | amritraj
 public | anomaly                 | table | amritraj
 public | anomaly_event           | table | amritraj
 public | anomaly_indicator       | table | amritraj
 public | anomaly_indicator_value | table | amritraj
 public | app_user                | table | amritraj
 public | company                 | table | amritraj
 public | event_news              | table | amritraj
 public | exchange                | table | amritraj
 public | market_data             | table | amritraj
 public | market_event            | table | amritraj
 public | news_article            | table | amritraj
 public | sector                  | table | amritraj
 public | stock                   | table | amritraj
 public | trading_day             | table | amritraj
 public | user_preference         | table | amritraj
 public | watchlist               | table | amritraj
 public | watchlist_item          | table | amritraj
(18 rows)
```

```
financial_market_ddl_commands=# \d sector
                                      Table "public.sector"
   Column    |          Type          | Collation | Nullable |                  Default
-------------+------------------------+-----------+----------+-----------------------------------
------
 sector_id   | integer                |           | not null | nextval('sector_sector_id_seq'::regc
lass)
 name        | character varying(100) |           | not null |
 description | text                   |           |          |
Indexes:
    "sector_pkey" PRIMARY KEY, btree (sector_id)
    "sector_name_key" UNIQUE CONSTRAINT, btree (name)
Referenced by:
    TABLE "company" CONSTRAINT "fk_company_sector" FOREIGN KEY (sector_id) REFERENCES sector(sector
_id)

financial_market_ddl_commands=# \d company
                                      Table "public.company"
   Column   |          Type          | Collation | Nullable |                  Default
------------+------------------------+-----------+----------+-----------------------------------
-------
 company_id | integer                |           | not null | nextval('company_company_id_seq'::reg
class)
 name       | character varying(150) |           | not null |
 sector_id  | integer                |           | not null |
 industry   | character varying(100) |           |          |
 country    | character varying(100) |           |          |
 website    | character varying(255) |           |          |
Indexes:
    "company_pkey" PRIMARY KEY, btree (company_id)
    "idx_company_sector" btree (sector_id)
Foreign-key constraints:
    "fk_company_sector" FOREIGN KEY (sector_id) REFERENCES sector(sector_id)
Referenced by:
    TABLE "stock" CONSTRAINT "fk_stock_company" FOREIGN KEY (company_id) REFERENCES company(company
_id)

financial_market_ddl_commands=# \d exchange
                                       Table "public.exchange"
   Column    |          Type          | Collation | Nullable |                   Default
-------------+------------------------+-----------+----------+-----------------------------------
----------
 exchange_id | integer                |           | not null | nextval('exchange_exchange_id_seq'::
regclass)
 code        | character varying(20)  |           | not null |
 name        | character varying(100) |           | not null |
 country     | character varying(100) |           |          |
 timezone    | character varying(50)  |           |          |
Indexes:
    "exchange_pkey" PRIMARY KEY, btree (exchange_id)
    "exchange_code_key" UNIQUE CONSTRAINT, btree (code)
Referenced by:
    TABLE "stock" CONSTRAINT "fk_stock_exchange" FOREIGN KEY (exchange_id) REFERENCES exchange(exch
ange_id)
    TABLE "trading_day" CONSTRAINT "fk_trading_day_exchange" FOREIGN KEY (exchange_id) REFERENCES e
xchange(exchange_id)
```

```
financial_market_ddl_commands=# \d stock
                                        Table "public.stock"
     Column      |         Type          | Collation | Nullable |                 Default
-----------------+-----------------------+-----------+----------+---------------------------------
-------
 stock_id        | integer               |           | not null | nextval('stock_stock_id_seq'::reg
class)
 exchange_id     | integer               |           | not null |
 company_id      | integer               |           | not null |
 ticker          | character varying(20) |           | not null |
 isin            | character varying(20) |           |          |
 instrument_type | character varying(50) |           |          |
 currency        | character varying(10) |           |          |
 is_active       | boolean               |           | not null | true
Indexes:
    "stock_pkey" PRIMARY KEY, btree (stock_id)
    "idx_stock_company" btree (company_id)
    "idx_stock_exchange" btree (exchange_id)
    "stock_isin_key" UNIQUE CONSTRAINT, btree (isin)
    "uq_stock_exchange_ticker" UNIQUE CONSTRAINT, btree (exchange_id, ticker)
Foreign-key constraints:
    "fk_stock_company" FOREIGN KEY (company_id) REFERENCES company(company_id)
    "fk_stock_exchange" FOREIGN KEY (exchange_id) REFERENCES exchange(exchange_id)
Referenced by:
    TABLE "market_data" CONSTRAINT "fk_market_data_stock" FOREIGN KEY (stock_id) REFERENCES stock(s
tock_id)
    TABLE "watchlist_item" CONSTRAINT "fk_watchlist_item_stock" FOREIGN KEY (stock_id) REFERENCES s
tock(stock_id) ON DELETE CASCADE

financial_market_ddl_commands=# \d trading_day
               Table "public.trading_day"
   Column    |  Type   | Collation | Nullable | Default
-------------+---------+-----------+----------+---------
 exchange_id | integer |           | not null |
 trade_date  | date    |           | not null |
 is_holiday  | boolean |           | not null | false
 description | text    |           |          |
Indexes:
    "pk_trading_day" PRIMARY KEY, btree (exchange_id, trade_date)
    "idx_trading_day_date" btree (trade_date)
Foreign-key constraints:
    "fk_trading_day_exchange" FOREIGN KEY (exchange_id) REFERENCES exchange(exchange_id)
Referenced by:
    TABLE "market_data" CONSTRAINT "fk_market_data_trading_day" FOREIGN KEY (exchange_id, trade_dat
e) REFERENCES trading_day(exchange_id, trade_date)

financial_market_ddl_commands=# \d market_data
                                      Table "public.market_data"
     Column     |     Type      | Collation | Nullable |                       Default
----------------+---------------+-----------+----------+---------------------------------------
----------
 market_data_id | bigint        |           | not null | nextval('market_data_market_data_id_seq'::
regclass)
 stock_id       | integer       |           | not null |
 exchange_id    | integer       |           | not null |
 trade_date     | date          |           | not null |
 open_price     | numeric(18,6) |           |          |
 high_price     | numeric(18,6) |           |          |
 low_price      | numeric(18,6) |           |          |
 close_price    | numeric(18,6) |           |          |
 adjusted_close | numeric(18,6) |           |          |
 volume         | bigint        |           |          |
 turnover       | numeric(20,4) |           |          |
Indexes:
    "market_data_pkey" PRIMARY KEY, btree (market_data_id)
    "idx_market_data_stock_date" btree (stock_id, trade_date)
    "idx_market_data_trade_date" btree (trade_date)
    "uq_market_data_stock_date" UNIQUE CONSTRAINT, btree (stock_id, trade_date)
Check constraints:
    "chk_market_data_adjusted_close" CHECK (adjusted_close IS NULL OR adjusted_close >= 0::numeric)
    "chk_market_data_close" CHECK (close_price IS NULL OR close_price >= 0::numeric)
    "chk_market_data_high" CHECK (high_price IS NULL OR high_price >= 0::numeric)
    "chk_market_data_high_low" CHECK (high_price IS NULL OR low_price IS NULL OR high_price >= low_
price)
    "chk_market_data_low" CHECK (low_price IS NULL OR low_price >= 0::numeric)
    "chk_market_data_prices" CHECK (open_price IS NULL OR open_price >= 0::numeric)
    "chk_market_data_turnover" CHECK (turnover IS NULL OR turnover >= 0::numeric)
    "chk_market_data_volume" CHECK (volume IS NULL OR volume >= 0)
Foreign-key constraints:
    "fk_market_data_stock" FOREIGN KEY (stock_id) REFERENCES stock(stock_id)
    "fk_market_data_trading_day" FOREIGN KEY (exchange_id, trade_date) REFERENCES trading_day(excha
nge_id, trade_date)
Referenced by:
    TABLE "anomaly" CONSTRAINT "fk_anomaly_market_data" FOREIGN KEY (market_data_id) REFERENCES mar
ket_data(market_data_id)
```

```
[financial_market_ddl_commands=# \d anomaly                                                                    ]
                                        Table "public.anomaly"
     Column     |            Type             | Collation | Nullable |                    Default

----------------+-----------------------------+-----------+----------+-----------------------------
---------------
 anomaly_id     | bigint                      |           | not null | nextval('anomaly_anomaly_id_
seq'::regclass)
 market_data_id | bigint                      |           | not null |
 anomaly_type   | character varying(100)      |           | not null |
 severity_level | character varying(20)       |           | not null |
 score          | numeric(10,4)               |           |          |
 description    | text                        |           |          |
 detected_at    | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
Indexes:
    "anomaly_pkey" PRIMARY KEY, btree (anomaly_id)
    "idx_anomaly_detected_at" btree (detected_at)
    "idx_anomaly_market_data" btree (market_data_id)
    "idx_anomaly_severity" btree (severity_level)
    "idx_anomaly_type" btree (anomaly_type)
Check constraints:
    "chk_anomaly_score" CHECK (score IS NULL OR score >= 0::numeric)
    "chk_anomaly_severity" CHECK (severity_level::text = ANY (ARRAY['LOW'::character varying, 'MEDI
UM'::character varying, 'HIGH'::character varying, 'CRITICAL'::character varying]::text[]))
Foreign-key constraints:
    "fk_anomaly_market_data" FOREIGN KEY (market_data_id) REFERENCES market_data(market_data_id)
Referenced by:
    TABLE "anomaly_indicator_value" CONSTRAINT "fk_aiv_anomaly" FOREIGN KEY (anomaly_id) REFERENCES
 anomaly(anomaly_id) ON DELETE CASCADE
    TABLE "alert" CONSTRAINT "fk_alert_anomaly" FOREIGN KEY (anomaly_id) REFERENCES anomaly(anomaly
_id) ON DELETE CASCADE
    TABLE "anomaly_event" CONSTRAINT "fk_anomaly_event_anomaly" FOREIGN KEY (anomaly_id) REFERENCES
 anomaly(anomaly_id) ON DELETE CASCADE

[financial_market_ddl_commands=# \d anomaly_indicator                                                          ]
                                      Table "public.anomaly_indicator"
    Column    |          Type          | Collation | Nullable |                          Default

--------------+------------------------+-----------+----------+-----------------------------------
---------------------
 indicator_id | integer                |           | not null | nextval('anomaly_indicator_indicato
r_id_seq'::regclass)
 name         | character varying(100) |           | not null |
 description  | text                   |           |          |
 unit         | character varying(50)  |           |          |
Indexes:
    "anomaly_indicator_pkey" PRIMARY KEY, btree (indicator_id)
    "anomaly_indicator_name_key" UNIQUE CONSTRAINT, btree (name)
Referenced by:
    TABLE "anomaly_indicator_value" CONSTRAINT "fk_aiv_indicator" FOREIGN KEY (indicator_id) REFERE
NCES anomaly_indicator(indicator_id)

[financial_market_ddl_commands=# \d anomaly_indicator_value                                                    ]
            Table "public.anomaly_indicator_value"
     Column      |     Type      | Collation | Nullable | Default
-----------------+---------------+-----------+----------+---------
 anomaly_id      | bigint        |           | not null |
 indicator_id    | integer       |           | not null |
 actual_value    | numeric(20,8) |           |          |
 threshold_value | numeric(20,8) |           |          |
 deviation       | numeric(20,8) |           |          |
Indexes:
    "pk_anomaly_indicator_value" PRIMARY KEY, btree (anomaly_id, indicator_id)
Foreign-key constraints:
    "fk_aiv_anomaly" FOREIGN KEY (anomaly_id) REFERENCES anomaly(anomaly_id) ON DELETE CASCADE
    "fk_aiv_indicator" FOREIGN KEY (indicator_id) REFERENCES anomaly_indicator(indicator_id)

[financial_market_ddl_commands=# \d market_event                                                               ]
                                       Table "public.market_event"
    Column    |            Type             | Collation | Nullable |                     Default

--------------+-----------------------------+-----------+----------+-------------------------------
-----------------
 event_id     | bigint                      |           | not null | nextval('market_event_event_id
_seq'::regclass)
 start_date   | date                        |           | not null |
 end_date     | date                        |           |          |
 event_type   | character varying(100)      |           | not null |
 title        | character varying(255)      |           | not null |
 description  | text                        |           |          |
 impact_level | character varying(20)       |           |          |
 created_at   | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
Indexes:
    "market_event_pkey" PRIMARY KEY, btree (event_id)
    "idx_market_event_dates" btree (start_date, end_date)
    "idx_market_event_type" btree (event_type)
Check constraints:
    "chk_market_event_dates" CHECK (end_date IS NULL OR end_date >= start_date)
    "chk_market_event_impact" CHECK (impact_level IS NULL OR (impact_level::text = ANY (ARRAY['LOW'
::character varying, 'MEDIUM'::character varying, 'HIGH'::character varying, 'CRITICAL'::character
varying]::text[])))
Referenced by:
    TABLE "anomaly_event" CONSTRAINT "fk_anomaly_event_event" FOREIGN KEY (event_id) REFERENCES mar
ket_event(event_id) ON DELETE CASCADE
    TABLE "event_news" CONSTRAINT "fk_event_news_event" FOREIGN KEY (event_id) REFERENCES market_ev
ent(event_id) ON DELETE CASCADE
```

```
financial_market_ddl_commands=# \d anomaly_event
            Table "public.anomaly_event"
   Column   |  Type  | Collation | Nullable | Default
------------+--------+-----------+----------+---------
 anomaly_id | bigint |           | not null |
 event_id   | bigint |           | not null |
Indexes:
    "pk_anomaly_event" PRIMARY KEY, btree (anomaly_id, event_id)
Foreign-key constraints:
    "fk_anomaly_event_anomaly" FOREIGN KEY (anomaly_id) REFERENCES anomaly(anomaly_id) ON DELETE CA
SCADE
    "fk_anomaly_event_event" FOREIGN KEY (event_id) REFERENCES market_event(event_id) ON DELETE CAS
CADE

financial_market_ddl_commands=# \d news_article
                                      Table "public.news_article"
    Column    |            Type             | Collation | Nullable |                     Default

--------------+-----------------------------+-----------+----------+-------------------------------
------------------
 article_id   | bigint                      |           | not null | nextval('news_article_article_
id_seq'::regclass)
 source       | character varying(150)      |           | not null |
 title        | character varying(500)      |           | not null |
 article_url  | character varying(1000)     |           |          |
 published_at | timestamp without time zone |           |          |
 content      | text                        |           |          |
 embedding    | vector(384)                 |           |          |
Indexes:
    "news_article_pkey" PRIMARY KEY, btree (article_id)
    "idx_news_embedding" hnsw (embedding vector_cosine_ops)
    "idx_news_published_at" btree (published_at)
    "idx_news_source" btree (source)
    "news_article_article_url_key" UNIQUE CONSTRAINT, btree (article_url)
Referenced by:
    TABLE "event_news" CONSTRAINT "fk_event_news_article" FOREIGN KEY (article_id) REFERENCES news_
article(article_id) ON DELETE CASCADE

financial_market_ddl_commands=# \d event_news
                  Table "public.event_news"
     Column      |     Type     | Collation | Nullable | Default
-----------------+--------------+-----------+----------+---------
 event_id        | bigint       |           | not null |
 article_id      | bigint       |           | not null |
 relevance_score | numeric(6,4) |           |          |
Indexes:
    "pk_event_news" PRIMARY KEY, btree (event_id, article_id)
Check constraints:
    "chk_event_news_relevance" CHECK (relevance_score IS NULL OR relevance_score >= 0::numeric AND
relevance_score <= 1::numeric)
Foreign-key constraints:
    "fk_event_news_article" FOREIGN KEY (article_id) REFERENCES news_article(article_id) ON DELETE
CASCADE
    "fk_event_news_event" FOREIGN KEY (event_id) REFERENCES market_event(event_id) ON DELETE CASCAD
E
financial_market_ddl_commands=# \d app_user
                                     Table "public.app_user"
    Column     |            Type             | Collation | Nullable |                  Default

---------------+-----------------------------+-----------+----------+-----------------------------
-------------
 user_id       | bigint                      |           | not null | nextval('app_user_user_id_seq
'::regclass)
 name          | character varying(100)      |           | not null |
 email         | character varying(255)      |           | not null |
 password_hash | character varying(255)      |           | not null |
 role          | character varying(30)       |           | not null |
 created_at    | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 is_active     | boolean                     |           | not null | true
Indexes:
    "app_user_pkey" PRIMARY KEY, btree (user_id)
    "app_user_email_key" UNIQUE CONSTRAINT, btree (email)
Referenced by:
    TABLE "alert" CONSTRAINT "fk_alert_user" FOREIGN KEY (user_id) REFERENCES app_user(user_id) ON
DELETE CASCADE
    TABLE "user_preference" CONSTRAINT "fk_user_preference_user" FOREIGN KEY (user_id) REFERENCES a
pp_user(user_id) ON DELETE CASCADE
    TABLE "watchlist" CONSTRAINT "fk_watchlist_user" FOREIGN KEY (user_id) REFERENCES app_user(user
_id) ON DELETE CASCADE

financial_market_ddl_commands=# \d watchlist
                                      Table "public.watchlist"
    Column    |            Type             | Collation | Nullable |                     Default

--------------+-----------------------------+-----------+----------+-------------------------------
------------------
 watchlist_id | bigint                      |           | not null | nextval('watchlist_watchlist_i
d_seq'::regclass)
 user_id      | bigint                      |           | not null |
 name         | character varying(100)      |           | not null |
 created_at   | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
Indexes:
    "watchlist_pkey" PRIMARY KEY, btree (watchlist_id)
    "idx_watchlist_user" btree (user_id)
    "uq_watchlist_user_name" UNIQUE CONSTRAINT, btree (user_id, name)
Foreign-key constraints:
    "fk_watchlist_user" FOREIGN KEY (user_id) REFERENCES app_user(user_id) ON DELETE CASCADE
Referenced by:
    TABLE "user_preference" CONSTRAINT "fk_user_preference_watchlist" FOREIGN KEY (watchlist_id) RE
FERENCES watchlist(watchlist_id) ON DELETE SET NULL
    TABLE "watchlist_item" CONSTRAINT "fk_watchlist_item_watchlist" FOREIGN KEY (watchlist_id) REFE
RENCES watchlist(watchlist_id) ON DELETE CASCADE
```

```
financial_market_ddl_commands=# \d watchlist_item
                          Table "public.watchlist_item"
    Column    |            Type             | Collation | Nullable |      Default
--------------+-----------------------------+-----------+----------+-------------------
 watchlist_id | bigint                      |           | not null |
 stock_id     | integer                     |           | not null |
 added_at     | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
Indexes:
    "pk_watchlist_item" PRIMARY KEY, btree (watchlist_id, stock_id)
    "idx_watchlist_item_stock" btree (stock_id)
Foreign-key constraints:
    "fk_watchlist_item_stock" FOREIGN KEY (stock_id) REFERENCES stock(stock_id) ON DELETE CASCADE
    "fk_watchlist_item_watchlist" FOREIGN KEY (watchlist_id) REFERENCES watchlist(watchlist_id) ON
DELETE CASCADE

financial_market_ddl_commands=# \d user_preference
                                      Table "public.user_preference"
       Column        |         Type          | Collation | Nullable |                          Defaul
t
---------------------+-----------------------+-----------+----------+-----------------------------
-------------------------
 preference_id       | bigint                |           | not null | nextval('user_preference_pref
erence_id_seq'::regclass)
 user_id             | bigint                |           | not null |
 watchlist_id        | bigint                |           |          |
 alert_type          | character varying(50) |           |          |
 notification_method | character varying(50) |           |          |
 is_active           | boolean               |           | not null | true
Indexes:
    "user_preference_pkey" PRIMARY KEY, btree (preference_id)
Foreign-key constraints:
    "fk_user_preference_user" FOREIGN KEY (user_id) REFERENCES app_user(user_id) ON DELETE CASCADE
    "fk_user_preference_watchlist" FOREIGN KEY (watchlist_id) REFERENCES watchlist(watchlist_id) ON
 DELETE SET NULL
```

```
financial_market_ddl_commands=# \d alert
                                    Table "public.alert"
   Column   |            Type             | Collation | Nullable |                  Default

------------+-----------------------------+-----------+----------+---------------------------------
--------
 alert_id   | bigint                      |           | not null | nextval('alert_alert_id_seq'::re
gclass)
 anomaly_id | bigint                      |           | not null |
 user_id    | bigint                      |           | not null |
 alert_type | character varying(50)       |           |          |
 message    | text                        |           |          |
 is_read    | boolean                     |           | not null | false
 created_at | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
Indexes:
    "alert_pkey" PRIMARY KEY, btree (alert_id)
    "idx_alert_anomaly" btree (anomaly_id)
    "idx_alert_user" btree (user_id)
Foreign-key constraints:
    "fk_alert_anomaly" FOREIGN KEY (anomaly_id) REFERENCES anomaly(anomaly_id) ON DELETE CASCADE
    "fk_alert_user" FOREIGN KEY (user_id) REFERENCES app_user(user_id) ON DELETE CASCADE
```

```
financial_market_ddl_commands=# \di
                              List of relations
 Schema |             Name              | Type  |  Owner   |          Table
--------+-------------------------------+-------+----------+-------------------------
 public | alert_pkey                    | index | amritraj | alert
 public | anomaly_indicator_name_key    | index | amritraj | anomaly_indicator
 public | anomaly_indicator_pkey        | index | amritraj | anomaly_indicator
 public | anomaly_pkey                  | index | amritraj | anomaly
 public | app_user_email_key            | index | amritraj | app_user
 public | app_user_pkey                 | index | amritraj | app_user
 public | company_pkey                  | index | amritraj | company
 public | exchange_code_key             | index | amritraj | exchange
 public | exchange_pkey                 | index | amritraj | exchange
 public | idx_alert_anomaly             | index | amritraj | alert
 public | idx_alert_user                | index | amritraj | alert
 public | idx_anomaly_detected_at       | index | amritraj | anomaly
 public | idx_anomaly_market_data       | index | amritraj | anomaly
 public | idx_anomaly_severity          | index | amritraj | anomaly
 public | idx_anomaly_type              | index | amritraj | anomaly
 public | idx_company_sector            | index | amritraj | company
 public | idx_market_data_stock_date    | index | amritraj | market_data
 public | idx_market_data_trade_date    | index | amritraj | market_data
 public | idx_market_event_dates        | index | amritraj | market_event
 public | idx_market_event_type         | index | amritraj | market_event
 public | idx_news_embedding            | index | amritraj | news_article
 public | idx_news_published_at         | index | amritraj | news_article
 public | idx_news_source               | index | amritraj | news_article
 public | idx_stock_company             | index | amritraj | stock
 public | idx_stock_exchange            | index | amritraj | stock
 public | idx_trading_day_date          | index | amritraj | trading_day
 public | idx_watchlist_item_stock      | index | amritraj | watchlist_item
 public | idx_watchlist_user            | index | amritraj | watchlist
 public | market_data_pkey              | index | amritraj | market_data
 public | market_event_pkey             | index | amritraj | market_event
 public | news_article_article_url_key  | index | amritraj | news_article
 public | news_article_pkey             | index | amritraj | news_article
 public | pk_anomaly_event              | index | amritraj | anomaly_event
 public | pk_anomaly_indicator_value    | index | amritraj | anomaly_indicator_value
 public | pk_event_news                 | index | amritraj | event_news
 public | pk_trading_day                | index | amritraj | trading_day
 public | pk_watchlist_item             | index | amritraj | watchlist_item
 public | sector_name_key               | index | amritraj | sector
 public | sector_pkey                   | index | amritraj | sector
 public | stock_isin_key                | index | amritraj | stock
 public | stock_pkey                    | index | amritraj | stock
 public | uq_market_data_stock_date     | index | amritraj | market_data
 public | uq_stock_exchange_ticker      | index | amritraj | stock
 public | uq_watchlist_user_name        | index | amritraj | watchlist
 public | user_preference_pkey          | index | amritraj | user_preference
 public | watchlist_pkey                | index | amritraj | watchlist
(46 rows)

financial_market_ddl_commands=# \d news_article
                                        Table "public.news_article"
    Column    |            Type             | Collation | Nullable |                      Default
--------------+-----------------------------+-----------+----------+-------------------------------
------------------
 article_id   | bigint                      |           | not null | nextval('news_article_article_
id_seq'::regclass)
 source       | character varying(150)      |           | not null |
 title        | character varying(500)      |           | not null |
 article_url  | character varying(1000)     |           |          |
 published_at | timestamp without time zone |           |          |
 content      | text                        |           |          |
 embedding    | vector(384)                 |           |          |
Indexes:
    "news_article_pkey" PRIMARY KEY, btree (article_id)
    "idx_news_embedding" hnsw (embedding vector_cosine_ops)
    "idx_news_published_at" btree (published_at)
    "idx_news_source" btree (source)
    "news_article_article_url_key" UNIQUE CONSTRAINT, btree (article_url)
Referenced by:
    TABLE "event_news" CONSTRAINT "fk_event_news_article" FOREIGN KEY (article_id) REFERENCES news_
article(article_id) ON DELETE CASCADE

financial_market_ddl_commands=# SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'news_article';
          indexname           |                                          indexdef

------------------------------+-----------------------------------------------------------------------
-------------------------------
 news_article_pkey            | CREATE UNIQUE INDEX news_article_pkey ON public.news_article USING
btree (article_id)
 news_article_article_url_key | CREATE UNIQUE INDEX news_article_article_url_key ON public.news_art
icle USING btree (article_url)
 idx_news_published_at        | CREATE INDEX idx_news_published_at ON public.news_article USING btr
ee (published_at)
 idx_news_source              | CREATE INDEX idx_news_source ON public.news_article USING btree (so
urce)
 idx_news_embedding           | CREATE INDEX idx_news_embedding ON public.news_article USING hnsw (
embedding vector_cosine_ops)
(5 rows)
```

```
financial_market_ddl_commands=# SELECT
    tc.table_name,
    tc.constraint_name,
    tc.constraint_type
FROM information_schema.table_constraints tc
WHERE tc.table_schema = 'public'
ORDER BY tc.table_name, tc.constraint_type, tc.constraint_name;
       table_name        |        constraint_name         | constraint_type
-------------------------+--------------------------------+-----------------
 alert                   | 2200_18551_1_not_null          | CHECK
 alert                   | 2200_18551_2_not_null          | CHECK
 alert                   | 2200_18551_3_not_null          | CHECK
 alert                   | 2200_18551_6_not_null          | CHECK
 alert                   | 2200_18551_7_not_null          | CHECK
 alert                   | fk_alert_anomaly               | FOREIGN KEY
 alert                   | fk_alert_user                  | FOREIGN KEY
 alert                   | alert_pkey                     | PRIMARY KEY
 anomaly                 | 2200_18403_1_not_null          | CHECK
 anomaly                 | 2200_18403_2_not_null          | CHECK
 anomaly                 | 2200_18403_3_not_null          | CHECK
 anomaly                 | 2200_18403_4_not_null          | CHECK
 anomaly                 | 2200_18403_7_not_null          | CHECK
 anomaly                 | chk_anomaly_score              | CHECK
 anomaly                 | chk_anomaly_severity           | CHECK
 anomaly                 | fk_anomaly_market_data         | FOREIGN KEY
 anomaly                 | anomaly_pkey                   | PRIMARY KEY
 anomaly_event           | 2200_18446_1_not_null          | CHECK
 anomaly_event           | 2200_18446_2_not_null          | CHECK
 anomaly_event           | fk_anomaly_event_anomaly       | FOREIGN KEY
 anomaly_event           | fk_anomaly_event_event         | FOREIGN KEY
 anomaly_event           | pk_anomaly_event               | PRIMARY KEY
 anomaly_indicator       | 2200_18392_1_not_null          | CHECK
 anomaly_indicator       | 2200_18392_2_not_null          | CHECK
 anomaly_indicator       | anomaly_indicator_pkey         | PRIMARY KEY
 anomaly_indicator       | anomaly_indicator_name_key     | UNIQUE
 anomaly_indicator_value | 2200_18419_1_not_null          | CHECK
 anomaly_indicator_value | 2200_18419_2_not_null          | CHECK
 anomaly_indicator_value | fk_aiv_anomaly                 | FOREIGN KEY
 anomaly_indicator_value | fk_aiv_indicator               | FOREIGN KEY
 anomaly_indicator_value | pk_anomaly_indicator_value     | PRIMARY KEY
 app_user                | 2200_18489_1_not_null          | CHECK
 app_user                | 2200_18489_2_not_null          | CHECK
 app_user                | 2200_18489_3_not_null          | CHECK
 app_user                | 2200_18489_4_not_null          | CHECK
 app_user                | 2200_18489_5_not_null          | CHECK
 app_user                | 2200_18489_6_not_null          | CHECK
 app_user                | 2200_18489_7_not_null          | CHECK
 app_user                | app_user_pkey                  | PRIMARY KEY
 app_user                | app_user_email_key             | UNIQUE
 company                 | 2200_18307_1_not_null          | CHECK
 company                 | 2200_18307_2_not_null          | CHECK
 company                 | 2200_18307_3_not_null          | CHECK
 company                 | fk_company_sector              | FOREIGN KEY
 company                 | company_pkey                   | PRIMARY KEY
 event_news              | 2200_18472_1_not_null          | CHECK
 event_news              | 2200_18472_2_not_null          | CHECK
 event_news              | chk_event_news_relevance       | CHECK
 event_news              | fk_event_news_article          | FOREIGN KEY
 event_news              | fk_event_news_event            | FOREIGN KEY
 event_news              | pk_event_news                  | PRIMARY KEY
```

market_data	chk_market_data_prices	CHECK
market_data	chk_market_data_turnover	CHECK
market_data	chk_market_data_volume	CHECK
market_data	fk_market_data_stock	FOREIGN KEY
market_data	fk_market_data_trading_day	FOREIGN KEY
market_data	market_data_pkey	PRIMARY KEY
market_data	uq_market_data_stock_date	UNIQUE
market_event	2200_18435_1_not_null	CHECK
market_event	2200_18435_2_not_null	CHECK
market_event	2200_18435_4_not_null	CHECK
market_event	2200_18435_5_not_null	CHECK
market_event	2200_18435_8_not_null	CHECK
market_event	chk_market_event_dates	CHECK
market_event	chk_market_event_impact	CHECK
market_event	market_event_pkey	PRIMARY KEY
news_article	2200_18462_1_not_null	CHECK
news_article	2200_18462_2_not_null	CHECK
news_article	2200_18462_3_not_null	CHECK
news_article	news_article_pkey	PRIMARY KEY
news_article	news_article_article_url_key	UNIQUE
sector	2200_18296_1_not_null	CHECK
sector	2200_18296_2_not_null	CHECK
sector	sector_pkey	PRIMARY KEY
sector	sector_name_key	UNIQUE
stock	2200_18330_1_not_null	CHECK
stock	2200_18330_2_not_null	CHECK
stock	2200_18330_3_not_null	CHECK
stock	2200_18330_4_not_null	CHECK
stock	2200_18330_8_not_null	CHECK
stock	fk_stock_company	FOREIGN KEY
stock	fk_stock_exchange	FOREIGN KEY
stock	stock_pkey	PRIMARY KEY
stock	stock_isin_key	UNIQUE
stock	uq_stock_exchange_ticker	UNIQUE
trading_day	2200_18351_1_not_null	CHECK
trading_day	2200_18351_2_not_null	CHECK
trading_day	2200_18351_3_not_null	CHECK
trading_day	fk_trading_day_exchange	FOREIGN KEY
trading_day	pk_trading_day	PRIMARY KEY
user_preference	2200_18533_1_not_null	CHECK
user_preference	2200_18533_2_not_null	CHECK
user_preference	2200_18533_6_not_null	CHECK
user_preference	fk_user_preference_user	FOREIGN KEY
user_preference	fk_user_preference_watchlist	FOREIGN KEY
user_preference	user_preference_pkey	PRIMARY KEY
watchlist	2200_18502_1_not_null	CHECK
watchlist	2200_18502_2_not_null	CHECK
watchlist	2200_18502_3_not_null	CHECK
watchlist	2200_18502_4_not_null	CHECK
watchlist	fk_watchlist_user	FOREIGN KEY
watchlist	watchlist_pkey	PRIMARY KEY
watchlist	uq_watchlist_user_name	UNIQUE
watchlist_item	2200_18516_1_not_null	CHECK
watchlist_item	2200_18516_2_not_null	CHECK
watchlist_item	2200_18516_3_not_null	CHECK
watchlist_item	fk_watchlist_item_stock	FOREIGN KEY
watchlist_item	fk_watchlist_item_watchlist	FOREIGN KEY
watchlist_item	pk_watchlist_item	PRIMARY KEY

(123 rows)

```
financial_market_ddl_commands=# SELECT
    tc.table_name,
    kcu.column_name,
    ccu.table_name AS referenced_table,
    ccu.column_name AS referenced_column
FROM information_schema.table_constraints AS tc
JOIN information_schema.key_column_usage AS kcu
    ON tc.constraint_name = kcu.constraint_name
    AND tc.table_schema = kcu.table_schema
JOIN information_schema.constraint_column_usage AS ccu
    ON ccu.constraint_name = tc.constraint_name
    AND ccu.table_schema = tc.table_schema
WHERE tc.constraint_type = 'FOREIGN KEY'
ORDER BY tc.table_name, kcu.column_name;
       table_name        |  column_name   | referenced_table  | referenced_column
-------------------------+----------------+-------------------+-------------------
 alert                   | anomaly_id     | anomaly           | anomaly_id
 alert                   | user_id        | app_user          | user_id
 anomaly                 | market_data_id | market_data       | market_data_id
 anomaly_event           | anomaly_id     | anomaly           | anomaly_id
 anomaly_event           | event_id       | market_event      | event_id
 anomaly_indicator_value | anomaly_id     | anomaly           | anomaly_id
 anomaly_indicator_value | indicator_id   | anomaly_indicator | indicator_id
 company                 | sector_id      | sector            | sector_id
 event_news              | article_id     | news_article      | article_id
 event_news              | event_id       | market_event      | event_id
 market_data             | exchange_id    | trading_day       | trade_date
 market_data             | exchange_id    | trading_day       | exchange_id
 market_data             | stock_id       | stock             | stock_id
 market_data             | trade_date     | trading_day       | exchange_id
 market_data             | trade_date     | trading_day       | trade_date
 stock                   | company_id     | company           | company_id
 stock                   | exchange_id    | exchange          | exchange_id
 trading_day             | exchange_id    | exchange          | exchange_id
 user_preference         | user_id        | app_user          | user_id
 user_preference         | watchlist_id   | watchlist         | watchlist_id
 watchlist               | user_id        | app_user          | user_id
 watchlist_item          | stock_id       | stock             | stock_id
 watchlist_item          | watchlist_id   | watchlist         | watchlist_id
(23 rows)
```

```
financial_market_ddl_commands=# \dv
Did not find any relations.
financial_market_ddl_commands=# SELECT COUNT(*)
FROM information_schema.tables
WHERE table_schema = 'public';
 count
-------
    18
(1 row)

financial_market_ddl_commands=# SELECT extname, extversion
FROM pg_extension
WHERE extname = 'vector';
 extname | extversion
---------+------------
 vector  | 0.8.6
(1 row)

financial_market_ddl_commands=# \q
amritraj@Amrits-Laptop Team Project %
```